



Leveraging Deep Learning and Computer Vision to Optimize Circular Economy and Industrial Recycling

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Entrepreneurship Capstone Project: MVP Development

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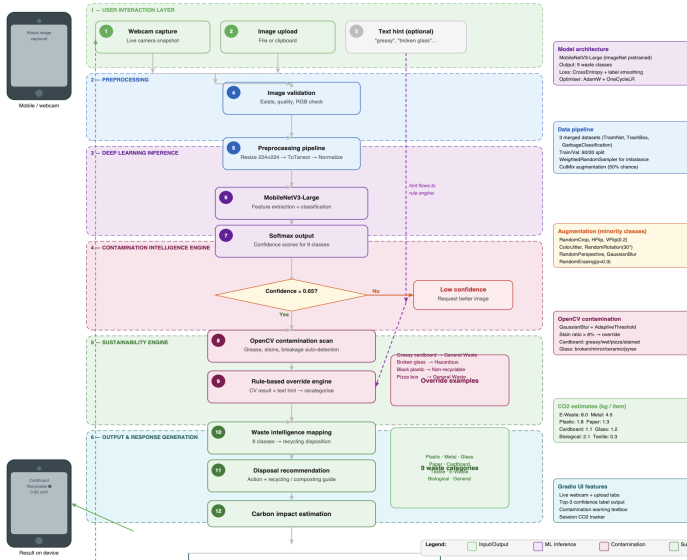
The Aim of This Project

To develop a scalable AI-driven 'Waste Intelligence System' that utilizes Deep Learning to automate material classification, aiming to reduce landfill contamination and enhance the efficiency of the circular economy.

Working Principle of AI Waste Intelligence System

Application Working Principle

AI Waste Intelligence System — MobileNetV3-Large + OpenCV Contamination Detection



Project Working Modules

Main Modules of the AI Waste Intelligence System

- Multi-Dataset Waste Image Collection and Merging
- Data Preprocessing and Label Mapping
- Train-Validation Dataset Splitting
- Image Augmentation using OpenCV and Torchvision
- Deep Learning Model Training using MobileNetV3
- Smart Waste Classification using AI Inference
- Contamination Detection using Computer Vision Rules
- Sustainability Impact and CO₂ Estimation
- Real-Time Waste Prediction using Webcam and Upload
- Interactive Gradio-based User Interface Deployment

Project Steps

- Dataset Preparation Module
 - Multi-Dataset Waste Image Collection ✓
 - Label Mapping and Data Merging ✓
 - Train-Validation Dataset Splitting ✓
 - Class Balancing using Weighted Sampling ✓
- Deep Learning Training Module
 - Image Preprocessing and Augmentation ✓
 - MobileNetV3-Large Model Implementation ✓
 - Model Training using Transfer Learning ✓
 - Performance Evaluation and Confusion Matrix ✓
- Smart Waste Intelligence Module
 - Real-Time Waste Classification ✓
 - Contamination Detection using Computer Vision ✓
 - Sustainability and CO₂ Impact Estimation ✓
 - Interactive Gradio Web Interface Deployment ✓

Unified Waste Dataset Structure

```
[6] print("\n✅ DATA MERGED SUCCESSFULLY")
✓ 6s print(f"Total images processed: {img_id}")

Processing: dataset-resized

Processing: trashbox_limited

Processing: garbage_classification

✅ DATA MERGED SUCCESSFULLY
Total images processed: 27388

[D]
✓ 0s for cls in OUT_DIR.iterdir():
    print(cls.name, ":", len(list(cls.glob("*"))))

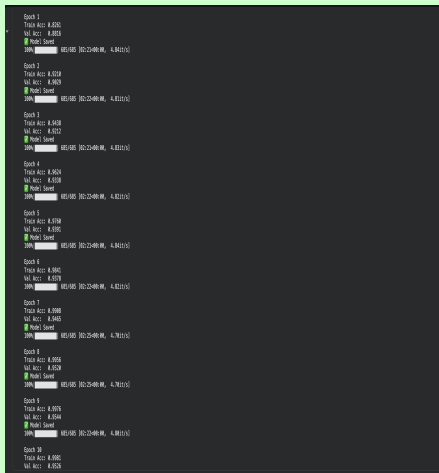
Plastic : 3196
Paper : 3788
Cardboard : 3391
General_Waste : 834
E-Waste : 945
Glass : 4647
Textile : 7302
Metal : 3285
```

- unified_dataset
 - Cardboard
 - E-Waste
 - General_Waste
 - Glass
 - Metal
 - Paper
 - Plastic
 - Textile

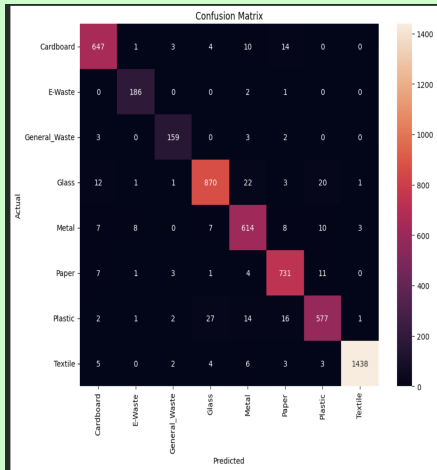
The waste datasets were merged into unified categories and balanced for AI model training. A total of 27,388 waste images were processed across multiple waste classes.

Model Training and Evaluation

Model Training Progress



Confusion Matrix Analysis



- Training Accuracy: 99.81%
- Validation Accuracy: 95.26%

- Multi-class waste classification

System Workflow Architecture

Complete Working Pipeline

Image Input → Preprocessing → CNN Model → Waste Classification

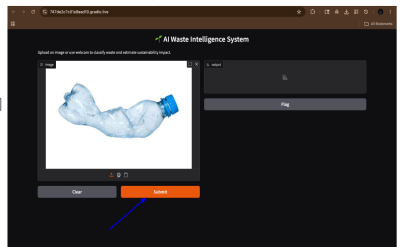
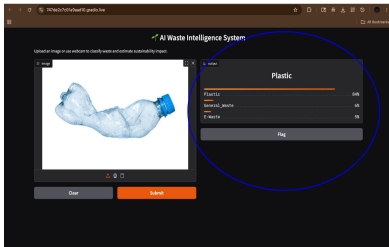
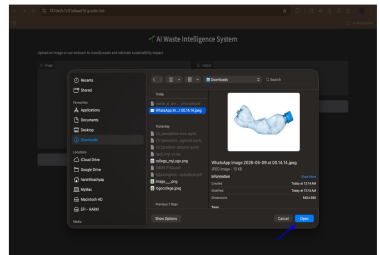
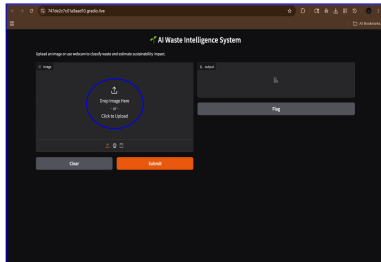
Input Stage

- Waste image captured
- Image resizing performed
- Noise reduction applied
- Data normalization

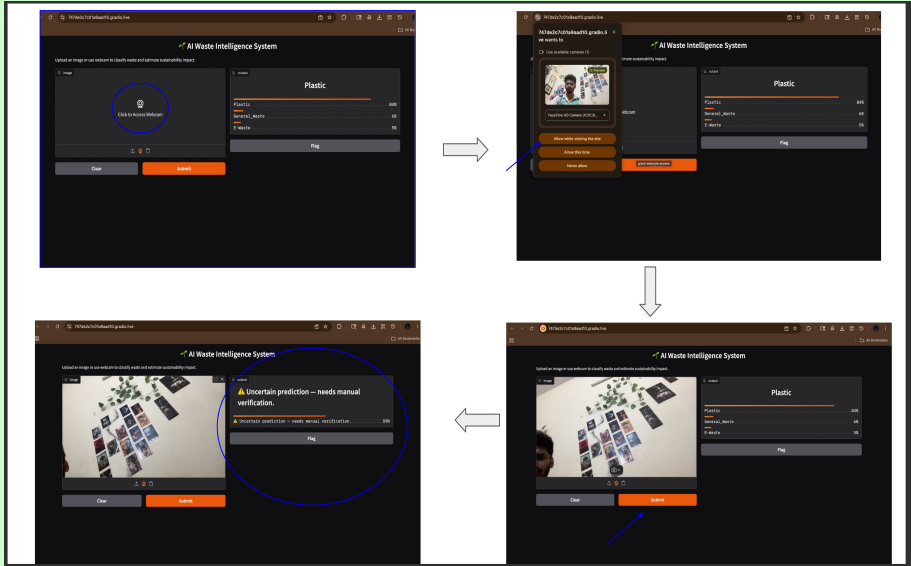
Prediction Stage

- CNN extracts features
- Waste category predicted
- Confidence score generated
- Result displayed to user

System Demonstration



Webcam-Based Waste Detection



Advantages of the Proposed System

Technical Advantages

- High Accuracy Waste Classification
- Real-Time Prediction System
- Multi-Class Waste Detection
- Webcam and Image Upload Support
- Automated Waste Recognition

Practical Advantages

- Supports Smart Waste Management
- Reduces Manual Sorting Effort
- Eco-Friendly Solution
- Easy to Use Web Interface
- Scalable for Smart City Applications

Applications of the System

Smart Waste Management

- Automated Waste Segregation
- Smart Recycling Systems
- Municipal Waste Monitoring
- Smart Dustbin Integration

Educational and Research Use

- AI and ML Research Projects
- Environmental Awareness Programs
- Smart Campus Waste Monitoring

Industrial Applications

- Recycling Industries
- E-Waste Processing Centers
- Waste Collection Automation

Future Smart City Use

- IoT-Based Waste Systems
- Real-Time Waste Analytics
- Sustainable City Development

Limitations of the System

- Accuracy depends on image quality
- Similar waste objects may cause misclassification
- Requires sufficient training data
- Real-time processing needs higher computational power
- Poor lighting conditions can affect predictions

Conclusion

Project Summary

The AI Waste Intelligence System successfully classifies multiple types of waste using Deep Learning and Computer Vision techniques.

The proposed system improves waste segregation efficiency, supports smart recycling processes, and contributes toward sustainable waste management solutions for smart cities and environmental protection.

- CNN-based Waste Classification
- Real-Time Prediction using Webcam
- Multi-Class Waste Detection
- User-Friendly Web Interface

References



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